



East West University
Department of Computer Science and Engineering

Course: CSE 475 (Machine Learning)
Section - 03

Submitted To :	Submitted By :
Dr. Md. Tauhid Bin Iqbal Assistant Professor Department of CSE East West University	Name : Hafsa Ferdousi Student ID : 2023-3-60-321 Department : CSE Date : 15 / 06 / 2026

Lab Report:03

Topic: Random Forest & Adaboost

Random Forest

n=5		n=10		n=15		n=20	
90:10	80:20	90:10	80:20	90:10	80:20	90:10	80:20
60	90	70	90	70	85	70	85

AdaBoost

n=5		n=10		n=15		n=20	
90:10	80:20	90:10	80:20	90:10	80:20	90:10	80:20
60	80	70	85	70	85	90	85

Analysis:

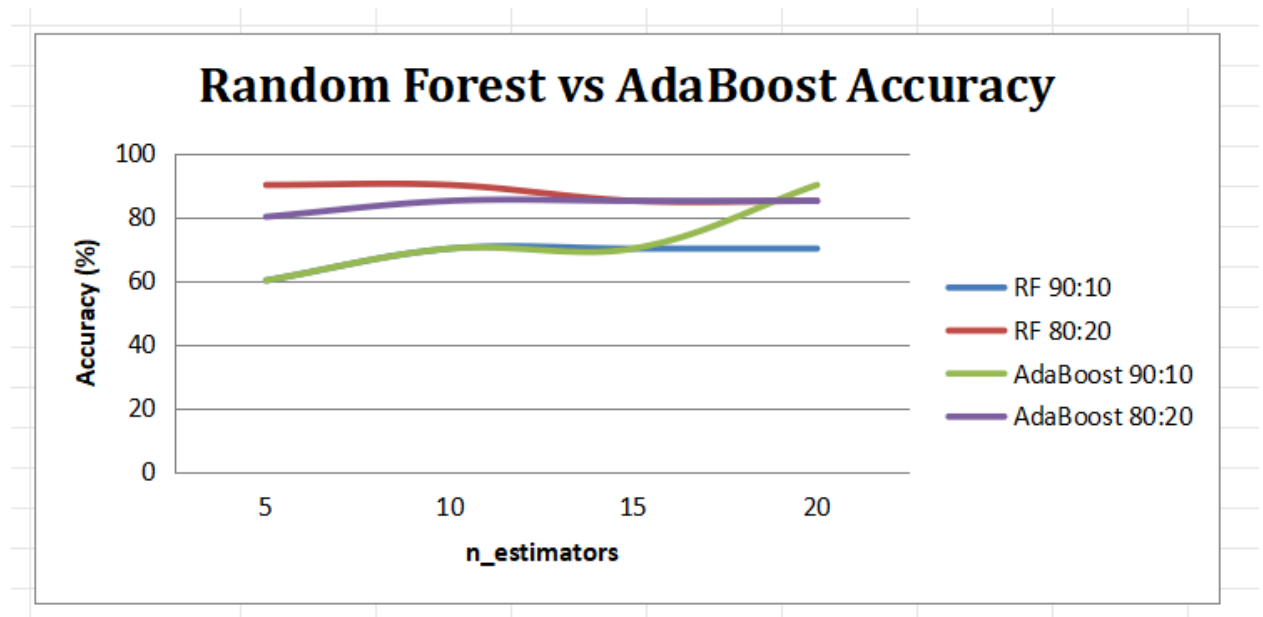


Figure: Random Forest & AdBoost Graph

Random Forest and AdaBoost were analyzed using different numbers of estimators and train-test split ratios. For the 90:10 split, Random Forest's accuracy increased from 60%

to 70% with 5 to 10 trees, stabilizing after 10 trees. Under the 80:20 split, Random Forest achieved a maximum accuracy of 90% with 5 and 10 trees, but performance declined to 85% with additional trees, suggesting that the model had learned most patterns by 10 trees. Conversely, AdaBoost saw gradual improvement from 60% to 90% across multiple estimators in the 90:10 split and stabilized at 85% in the 80:20 split, indicating that approximately 10 estimators are sufficient.

Interestingly, the 80:20 split outperformed the 90:10 split in specific instances, likely due to dataset size, the challenging nature of the samples in the test set, and random variations in splitting. For instance, with a small dataset of 100 samples, misclassifying one extra sample could drastically affect accuracy.

Both models achieved a peak accuracy of 90%, but Random Forest required fewer estimators compared to the 20 needed for AdaBoost, rendering it more efficient based on this analysis. In summary, Random Forest exhibited more stable performance with fewer estimators, highlighting its efficiency in achieving high accuracy.

Link:

 Random Forest & AdaBoost.ipynb